

Internet of Things

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Outline

- ▶ Data Handling and Analytics
- ▶ Qualitative & Quantitative Analysis
- ▶ Statistical models
- ▶ Analysis of variance
- ▶ Data dispersion
- ▶ Contingence and correlation
- ▶ Regression analysis
- ▶ Statistical significance
- ▶ Precision and Error limits
- ▶ Summary

What is Data Handling

▶ Data handling

- Ensures that research data is stored, archived or disposed off in a safe and secure manner during and after the conclusion of a research project.
- Includes the development of policies and procedures to manage data handled electronically as well as through non-electronic means

▶ In recent days, most data concern

➤ Big Data

- ✓ Due to heavy traffic generated by IoT devices
- ✓ Huge amount of data generated by the deployed sensors

What is Big Data

- ▶ “Big data technologies describe a new generation of technologies and architectures, designed to economically extract value from very large volumes of a wide variety of data, by enabling the high-velocity capture, discovery, and/or analysis.”

[Report of International Data Corporation (IDC)]

- ▶ “Big data shall mean the data of which the data volume, acquisition speed, or data representation limits the capacity of using traditional relational methods to conduct effective analysis or the data which may be effectively processed with important horizontal zoom technologies.”

[National Institute of Standards and Technology (NIST)]

Types of Data

➤ Structured data

Data that can be **easily organized**.

Usually stored in **relational databases**.

Structured Query Language (**SQL**) manages structured data in databases.

It accounts for only 20% of the total available data today in the world.

➤ Unstructured data

Information that **do not possess any pre-defined model**.

Traditional **RDBMSs are unable to process unstructured data**.

Enhances the ability to **provide better insight to huge datasets**.

It accounts for 80% of the total data available today in the world

Characteristics of Big Data

► Big Data is characterized by 7 Vs -

- ✓ Volume
- ✓ Velocity
- ✓ Variety
- ✓ Variability
- ✓ Veracity
- ✓ Visualization
- ✓ Value

Characteristics of Big Data

► Volume

- ✓ Quantity of data that is generated
- ✓ Sources of data are added continuously
- ✓ Example of volume-
 - ✓ 30TB of images will be generated every night from the Large Synoptic Survey Telescope (LSST)
 - ✓ 72 hours of video are uploaded to YouTube every minute

Characteristics of Big Data

Velocity

- ✓ Refers to the speed of generation of data
- ✓ Data processing time decreasing day-by-day in order to provide real-time services
- ✓ Older batch processing technology is unable to handle high velocity of data
- ✓ Example of velocity-
 - ✓ 140 million tweets per day on average (according to a survey conducted in 2011)
 - ✓ New York Stock Exchange captures 1TB of trade information during each trading session

Characteristics of Big Data (Contd.)

✓ **Variety**

- ✓ Refers to the category to which the data belongs
- ✓ No restriction over the input data formats
- ✓ Data mostly unstructured or semi-structured
- ✓ Example of variety-
 - ✓ Pure text, images, audio, video, web, GPS data, sensor data, SMS, documents, PDFs, flash etc.

Characteristics of Big Data

✓ Variability

- ✓ Refers to data whose meaning is **constantly changing**.
- ✓ Meaning of the data **depends on the context**.
- ✓ Data appear as an indecipherable mass without structure
- ✓ Example:
 - ✓ Language processing, Hashtags, Geo-spatial data, Multimedia, Sensor events

✓ Veracity

- ✓ Veracity refers to the **biases, noise and abnormality in data**.
- ✓ It is important in programs that involve **automated decision-making**, or **feeding the data**
- ✓ into **an unsupervised machine learning algorithm**.
- ✓ Veracity isn't just about data quality, it's about data **understandability**

Characteristics of Big Data

► Visualization

- ✓ Presentation of data in a **pictorial or graphical format**
- ✓ Enables decision makers to see analytics presented visually
- ✓ Identify new patterns

✓ Value

- ✓ It means extracting **useful business information from scattered data.**
- ✓ Includes a large **volume and variety of data**
- ✓ Easy to access and delivers quality analytics that enables informed decisions

Data Handling Technologies

- ✓ Cloud computing
 - ✓ Essential characteristics according to NIST
 - ✓ On-demand self service
 - ✓ Broad network access
 - ✓ Resource pooling
 - ✓ Rapid elasticity
 - ✓ Measured service
- ✓ Basic service models provided by cloud computing
 - ✓ Infrastructure-as-a-Service (IaaS)
 - ✓ Platform-as-a-Service (PaaS)
 - ✓ Software-as-a-Service (SaaS)

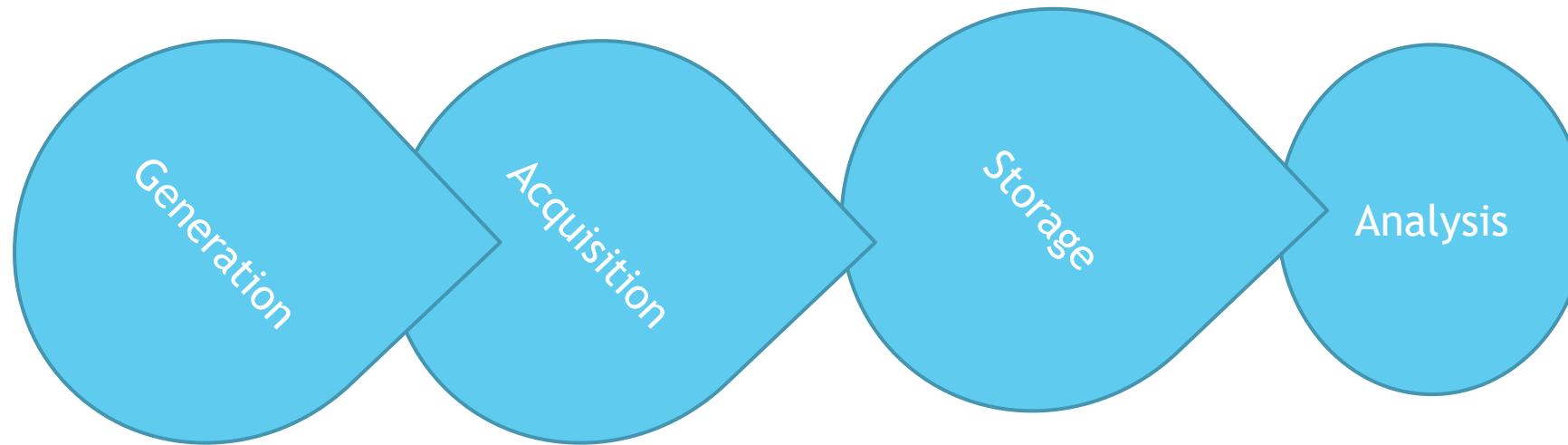
Data Handling Technologies

- ▶ Internet of Things (IoT)
 - ✓ According to Techopedia, IoT “in future where every day physical objects will be connected to the internet and will be able to identify themselves to other devices.”
 - ✓ Sensors embedded into various devices and machines and deployed into fields.
 - ✓ Sensors transmit sensed data to remote servers via Internet.
 - ✓ Continuous data acquisition from mobile equipment, transportation facilities, public facilities, and home appliances

Data Handling Technologies

- ✓ Data handling at data centers
 - ✓ Storing, managing, and organizing data.
 - ✓ Estimates and provides necessary processing capacity.
 - ✓ Provides sufficient network infrastructure.
 - ✓ Effectively manages energy consumption.
 - ✓ Replicates data to keep backup.
 - ✓ Develop business oriented strategic solutions from big data.
 - ✓ Helps business personnel to analyze existing data.
 - ✓ Discovers problems in business operations

Flow of Data



- ✓ Enterprise data
- ✓ IoT data
- ✓ Bio-medical data
- ✓ Other data

- ✓ Data collection
- ✓ Data transportation
- ✓ Data pre-processing

- ✓ Hadoop
- ✓ MapReduce
- ✓ NoSQL databases

- ✓ Bloom filter
- ✓ Parallel computing
- ✓ Hashing and indexing

Data Sources

- ✓ Enterprise data
 - ✓ Online trading and analysis data.
 - ✓ Production and inventory data.
 - ✓ Sales and other financial data.
- ✓ IoT data
 - ✓ Data from industry, agriculture, traffic, transportation
 - ✓ Medical-care data,
 - ✓ Data from public departments, and families.
- ✓ Bio-medical data
 - ✓ Masses of data generated by gene sequencing.
 - ✓ Data from medical clinics and medical R&Ds.
- ✓ Other fields
 - ✓ Fields such as - computational biology, astronomy, nuclear research etc

Data Acquisition

- ✓ Data collection
 - ▶ Log files or record files that are automatically generated by data sources to record activities for further analysis.
 - ▶ Sensory data such as sound wave, voice, vibration, automobile, chemical, current, weather, pressure, temperature etc.
 - ▶ Complex and variety of data collection through mobile devices. E.g. geographical location, 2D barcodes, pictures, videos etc.
- ✓ Data transmission
 - ✓ After collecting data, it will be transferred to storage system for further processing and analysis of the data.
 - ✓ Data transmission can be categorized as Inter-DCN transmission and Intra-DCN transmission.

Data Acquisition

- ✓ Data pre-processing
 - ✓ Collected datasets suffer from **noise, redundancy, inconsistency** etc., thus, pre-processing of data is necessary.
 - ✓ Pre-processing of relational data mainly follows -integration, cleaning, and redundancy mitigation
 - ✓ Integration is combining data from various sources and provides users with a uniform view of data.
 - ✓ Cleaning is identifying inaccurate, incomplete, or unreasonable data, and then modifying or deleting such data.
 - ✓ Redundancy mitigation is **eliminating data repetition through detection, filtering and compression of data to avoid unnecessary transmission.**

Data Storage

✓ File system

- ✓ Distributed file systems that store massive data and ensure -consistency, availability, and fault tolerance of data.
- ✓ GFS is a notable example of distributed file system that supports large-scale file system, though it's performance is limited in case of small files
- ✓ Hadoop Distributed File System (HDFS) and Kosmosfs are other notable file systems, derived from the open source codes of GFS.

✓ Databases

- ✓ Emergence of non-traditional relational databases (NoSQL) in order to deal with the characteristics that big data possess.
- ✓ Three main NoSQL databases -Key-value databases, column-oriented databases, and document-oriented databases.

Hadoop

- ✓ Hadoop is a software framework for distributed processing of large datasets across large clusters of computers.
- ✓ Hadoop is open-source implementation for Google 's GFS and MapReduce.
- ✓ Apache Hadoop's Map Reduce and Hadoop Distributed File System (HDFS) components originally derived respectively from Google's MapReduce and Google File System (GFS) .



Building Blocks of Hadoop

- ✓ Hadoop Common

- ✓ A module containing the utilities that support the other Hadoop components

Hadoop Distributed File System (HDFS)

- ✓ Provides reliable data storage and access across the nodes

- ✓ MapReduce

- ✓ Framework for applications that process large amount of datasets in parallel.

- ✓ Yet Another Resource Negotiator (YARN)

- ✓ Next-generation MapReduce, which assigns CPU, memory and storage to applications running on a Hadoop cluster.

Hadoop Distributed File System (HDFS)

► Centralized node

- ✓ Namenode
- ✓ Maintains metadata info about files

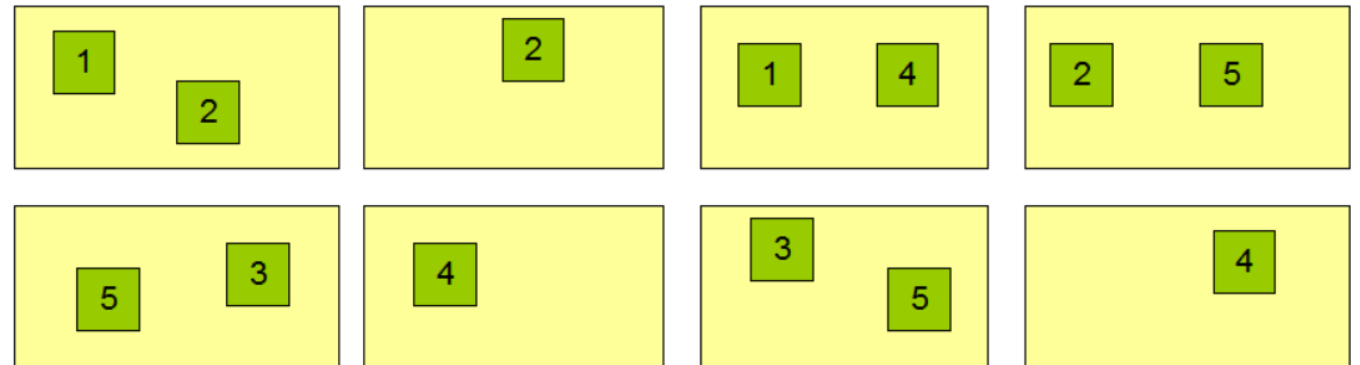
✓ Distributed node

- ✓ Datanode
- ✓ Store the actual data
- ✓ Files are divided into blocks
- ✓ Each block is replicate

Block Replication

Namenode (Filename, numReplicas, block-ids, ...)
/users/sameerp/data/part-0, r:2, {1,3}, ...
/users/sameerp/data/part-1, r:3, {2,4,5}, ...

Datanodes



Name and Data Nodes

- ✓ **Namenode**
 - ✓ Stores filesystem metadata.
 - ✓ Maintains two in-memory tables, to map the datanodes to the blocks, and vice versa
- ✓ **Datanode**
 - ✓ Stores actual data
 - ✓ Data nodes can talk to each other to rebalance and replicate data
 - ✓ Data nodes update the namenode with the block information periodically
 - ✓ Before updating datanodes verify the checksums.

Job and Task Trackers

- ✓ Job Tracker -
 - ✓ Runs with the Namenode
 - ✓ Receives the user's job
 - ✓ Decides on how many tasks will run (number of mappers)
 - ✓ Decides on where to run each mapper (concept of locality)
- ✓ Task Tracker -
 - ✓ Runs on each datanode
 - ✓ Receives the task from Job Tracker
 - ✓ Always in communication with the Job Tracker reporting progress

Hadoop Master/Slave Architecture

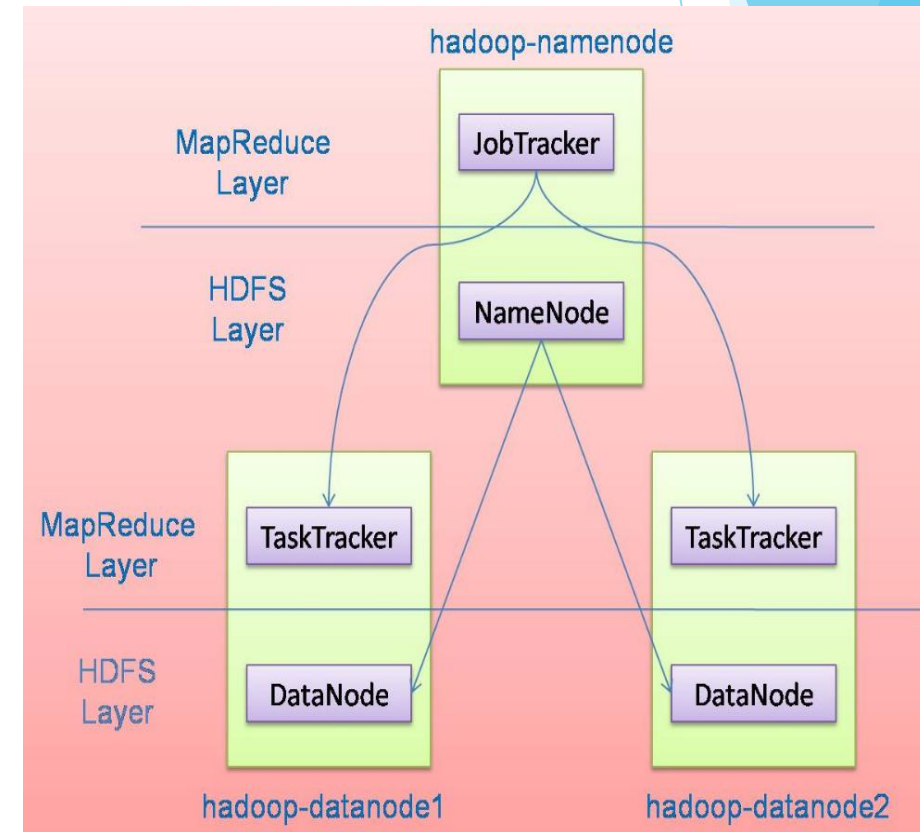
Master-slave shared-nothing architecture

- ✓ Master

- ✓ Executes operations like opening, closing, and renaming files and directories.
- ✓ Determines the mapping of blocks to Datanodes.

- ✓ Slave

- ✓ Serves read and write requests from the file system's clients.
- ✓ Performs block creation, deletion, and replication as instructed by the Namenode.



What is Data Analytics

“Data analytics (DA) is the process of examining data sets in order to draw conclusions about the information they contain, increasingly with the aid of specialized systems and software. Data analytics technologies and techniques are widely used in commercial industries to enable organizations to make more-informed business decisions and by scientists and researchers to verify or disprove scientific models, theories and hypotheses.”

Types of Data Analysis

- ✓ Two types of analysis
 - ✓ Qualitative Analysis
 - ✓ Deals with the analysis of data that is categorical in nature
 - ✓ Quantitative Analysis
 - ✓ Quantitative analysis refers to the process by which numerical data is analyzed

Qualitative Analysis

- ✓ Data is **not described through numerical values**
- ✓ Described by some sort of descriptive context such as text
- ✓ Data can be gathered by many methods such as interviews, videos and audio recordings, field notes
- ✓ Data needs to be interpreted
- ✓ The grouping of data into identifiable themes

Qualitative analysis can be summarized by three basic principles (Seidel, 1998):

- ✓ **Notice things**
- ✓ **Collect things**
- ✓ **Think about things**

Quantitative Analysis

- ▶ Quantitative analysis refers to the **process by which numerical data is analyzed**
- ▶ Involves descriptive statistics such as **mean, media, standard deviation**
- ▶ The following are often involved with quantitative analysis:

Statistical models

- ✓ Analysis of variables
 - ✓ Data dispersion
 - ✓ Analysis of relationships between variables
 - ✓ Contingence and correlation
-
- ✓ Regression analysis
 - ✓ Statistical significance
 - ✓ Precision
 - ✓ Error limits

Comparison

Qualitative Data	Quantitative Data
Data is observed	Data is measured
Involves descriptions	Involves numbers
Emphasis is on quality	Emphasis is on quantity
Examples are color, smell, taste, etc	Examples are volume, weight, etc.

Advantages

- ✓ Allows for the identification of important (and often mission-critical) trends
- ✓ Helps businesses identify performance problems that require some sort of action
- ✓ Can be viewed in a visual manner, which leads to faster and better decisions
- ✓ Better awareness regarding the habits of potential customers
- ✓ It can provide a company with an edge over their competitors

Statistical models

- ✓ The **statistical model** is defined as the mathematical equation that are formulated in the form of **relationships between variables**.
- ✓ A statistical model illustrates how **a set of random variables** is related to **another set of random variables**.
- ✓ A statistical model is represented as the **ordered pair** (X, P)
 - ✓ X denotes the set of all **possible observations**
 - ✓ P refers to the set of **probability distributions** on X

Statistical models (Contd.)

- ✓ Statistical models are broadly categorized as
 - ✓ Complete models
 - ✓ Incomplete models
- ✓ Complete model does have the number of variables equal to the number of equations
- ✓ An incomplete model does not have the same number of variables as the number of equations

Statistical models (Contd.)

- ✓ In order to build a statistical model
 - ✓ Data Gathering
 - ✓ Descriptive Methods
 - ✓ Thinking about Predictors
 - ✓ Building of model
 - ✓ Interpreting the Results

Analysis of variance

- ✓ Analysis of Variance (ANOVA) is a parametric statistical technique used to compare datasets.
- ✓ ANOVA is best applied where more than 2 populations or samples are meant to be compared.
- ✓ To perform an ANOVA, we must have a continuous response variable and at least one categorical factor (e.g. age, gender) with two or more levels (e.g. Locations 1, 2)
- ✓ ANOVAs require data from approximately normally distributed populations

Analysis of variance (Contd.)

- ✓ Properties to perform ANOVA -
- ✓ Independence of case
 - ✓ The sample should be selected randomly
 - ✓ There should not be any pattern in the selection of the sample
- ✓ Normality
- ✓ Distribution of each group should be normal
- ✓ Homogeneity
 - ✓ Variance between the groups should be the same (e.g. should not compare data from cities with those from slums)

Analysis of variance (Contd.)

- ✓ Analysis of variance (ANOVA) has three types:
- ✓ One way analysis
 - ✓ One fixed factor (levels set by investigator). Factors: age, gender, etc.
- ✓ Two way analysis
 - ✓ Factor variables are more than two
- ✓ K-way analysis
 - ✓ Factor variables are k

Analysis of variance (Contd.)

- ✓ **Total Sum of square**
 - ✓ In statistical data analysis, the total sum of squares (TSS or SST) is a quantity that appears as part of a standard way of presenting results of such analyses. It is defined as being the sum, over all observations, of the squared differences of each observation from the overall mean
- ✓ **F -ratio**
 - ✓ Helps to understand the ratio of variance between two data sets
 - ✓ The F ratio is approximately 1.0 when the null hypothesis is true and is greater than 1.0 when the null hypothesis is false.
- ✓ **Degree of freedom**
 - ✓ Factors which have no effect on the variance
 - ✓ The number of degrees of freedom is the number of values in the final calculation of a statistic that are free to vary

Data dispersion

- ✓ A measure of statistical dispersion is a nonnegative real number that is zero if all the data are the same and increases as the data becomes more diverse.
- ✓ Examples of dispersion measures:
 - ✓ Range
 - ✓ Average absolute deviation
 - ✓ Variance and Standard deviation

Data dispersion (Contd.)

- ✓ **Range**

- ✓ The range is calculated by simply taking the difference between the maximum and minimum values in the data set.

- ✓ **Average absolute deviation**

- ✓ The average absolute deviation (or mean absolute deviation) of a data set is the average of the absolute deviations from the mean.

- ✓ **Variance**

- ✓ Variance is the expectation of the squared deviation of a random variable from its mean

- ✓ **Standard deviation**

- ✓ Standard deviation (SD) is a measure that is used to quantify the amount of variation or dispersion of a set of data values

Contingence and correlation

- ▶ In statistics, a contingency table (also known as a **cross tabulation or crosstab**) is a type of table in a matrix format that displays the (multivariate) frequency distribution of the variables
- ▶ Provides a basic picture of the **interrelation between two variables**
- ▶ A crucial problem of multivariate statistics is finding (direct-)dependence structure underlying the variables contained in high-dimensional contingency tables

Contingence and correlation

- ▶ Correlation is a technique for investigating the relationship between two quantitative, continuous variables
- ▶ Pearson's correlation coefficient (r) is a measure of the strength of the association between the two variables.
- ▶ Correlations are useful because they can indicate a predictive relationship that can be exploited in practice

Regression analysis

- ▶ In statistical modeling, regression analysis is a statistical process for estimating the relationships among variables
- ▶ Focuses on the relationship between a dependent variable and one or more independent variables
- ▶ Regression analysis estimates the conditional expectation of the dependent variable given the independent variables

Regression analysis (Contd.)

- ▶ The estimation target is a function of the independent variables called the regression function
- ▶ Characterize the variation of the dependent variable around the regression function which can be described by a probability distribution
- ▶ Regression analysis is widely used for prediction and forecasting, where its use has substantial overlap with the field of machine learning
- ▶ Regression analysis is also used to understand which among the independent variables are related to the dependent variable

Statistical significance

- ✓ Statistical significance is the likelihood that the difference in conversion rates between a given variation and the baseline is not due to random chance
- ✓ Statistical significance level reflects the risk tolerance and confidence level
- ✓ There are two key variables that go into **determining statistical significance**:
 - ✓ Sample size
 - ✓ Effect size

Statistical significance (Contd.)

- ▶ Sample size refers to the sample size of the experiment
- ▶ The larger your sample size, the more confident you can be in the result of the experiment (assuming that it is a randomized sample)
- ▶ The effect size is just the standardized mean difference between the two groups
- ▶ If a particular experiment replicated, the different effect size estimates from each study can easily be combined to give an overall best estimate of the effect size

Precision and Error limits

- ▶ Precision refers to how close estimates from different samples are to each other
- ▶ The standard error is a measure of precision
- ▶ When the standard error is small, estimates from different samples will be close in value and vice versa
- ▶ Precision is inversely related to standard error

Precision and Error limits (Contd.)

- ✓ The limits of error are the maximum overestimate and the maximum underestimate from the combination of the sampling and the non-sampling errors
- ✓ The margin of error is defined as -
 - ✓ Limit of error = Critical value x Standard deviation of the statistic
 - ✓ Critical value: Determines the tolerance level of error.

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Question



Thank you

